Towards Sustainable and Resilient Federated Learning in the Edge-Cloud Continuum

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***Abstract*—Federated Learning (FL) is increasingly adopted in edge-cloud environments, where data locality, privacy constraints, intermittent connectivity, and heterogeneous device capabilities make centralized training impractical. However, the emerging edge-cloud continuum also exposes a less-studied systems problem: FL orchestration must simultaneously preserve predictive performance, reduce communication and energy consumption, account for variation in carbon intensity across locations, and remain robust to stragglers and failures. Existing work has made substantial progress in non-IID optimization, client selection, compression, and green computing, yet these dimensions are often evaluated in isolation. This paper proposes EcoRes-FL, a sustainability- and resilience-aware FL framework for the edge-cloud continuum. The framework combines carbon-aware client scoring, reliability filtering, proximal local optimization, and top-k sparse updates. We formulate client participation as a mixed-integer linear optimization problem that captures accuracy utility, communication cost, energy consumption, carbon intensity, latency deadlines, and minimum participation constraints. The preliminary results demonstrate that the proposed framework can substantially reduce communication and energy consumption while preserving competitive performance.**

***Index Terms*—Federated Learning, Edge-Cloud Continuum, Sustainable AI, Carbon-Aware Scheduling, Resilient Distributed Learning, Client Selection, Mixed-Integer Linear Programming.**

I. Introduction

The deployment of machine learning across connected devices has shifted from a cloud-only paradigm to a continuum spanning edge, fog, and cloud resources. Cameras, phones, vehicles, industrial controllers, medical devices, and smart meters now generate data near the physical processes they monitor. Moving all this data to a centralized cloud is often undesirable because of privacy, bandwidth, latency, governance, and cost constraints. Federated Learning (FL) addresses this tension by sending model-training code to the data rather than moving raw data to a centralized repository. The classical FedAvg algorithm demonstrated that decentralized model averaging can train deep models under unbalanced and non-IID data distributions while significantly reducing centralized data movement [1]. Production-scale FL systems later showed that practical deployment introduces additional constraints, including unreliable devices, partial participation, secure aggregation, scheduling, and orchestration [2].

The next generation of FL systems will not run solely on homogeneous mobile devices. They will execute across an edge-cloud continuum composed of constrained sensors, local gateways, micro-data centers, regional cloudlets, and high-capacity cloud servers. This architecture is attractive because it enables low-latency analytics close to the data source while preserving the elasticity of cloud resources. Yet the continuum also complicates FL orchestration. A client on a low-power edge node may have low carbon intensity because it is close to renewable generation, but it may be slow or unreliable. A cloud node may be fast and stable, but its operational carbon footprint can be high, depending on its energy mix and workload placement. Similarly, a client with statistically useful data may become a straggler under congestion, while a reliable client may contribute redundant or biased updates under non-IID partitioning. Therefore, sustainability, resilience, statistical utility, communication efficiency, and systems heterogeneity cannot be treated as isolated design goals.

The literature already provides important building blocks. FedAvg remains the canonical baseline for communication-efficient FL [1]. FedProx introduces a proximal term to stabilize optimization in the presence of statistical and system heterogeneity [3]. SCAFFOLD uses control variates to mitigate client drift under non-IID data [4]. Oort improves time-to-accuracy by selecting participants based on utility and system speed [5]. Gradient-compression techniques reduce communication volume and, consequently, energy use [6]. In parallel, carbon-aware scheduling and green AI research argue that energy and carbon reporting should be treated as first-class evaluation dimensions [7], [8]. Recent work, such as FedGreen and CAFE, has brought this discussion into the FL setting by adapting model sizes or selecting geographically distributed data-center participants under carbon budgets [9]–[11].

Despite these advances, a gap remains in edge-cloud FL: there is still limited reproducible work that jointly models (i) non-IID learning behavior, (ii) client-scale variation, (iii) device failures and stragglers, (iv) communication sparsifica-

tion, and (v) carbon-aware participation, all within a unified optimization framework. Several papers focus on accuracy and convergence; others emphasize communication; energy-efficient FL studies often focus on device energy without explicitly accounting for carbon intensity; and carbon-aware studies often assume data-center settings rather than mixed edge-cloud tiers. This fragmentation makes it difficult to answer practical questions such as: Which clients should participate in a given round when carbon intensity, bandwidth, reliability, and data utility vary? How much communication can be reduced before resilience or accuracy degrades? Under what level of non-IID severity does carbon-aware selection become harmful, neutral, or beneficial?

This paper develops a research framework to address these questions. We propose EcoRes-FL, a lightweight, sustainability- and resilience-aware FL orchestration method for the edge-cloud continuum. Rather than presenting a final production system, we contribute a reproducible evaluation framework that explicitly connects mathematical formulation, scheduling constraints, implementation, and visual analysis. The proposed method uses four mechanisms: carbon-aware client scoring, reliability-aware filtering, proximal local training, and top-k sparse model updates. The client-selection problem is formulated as a mixed-integer linear program (MILP), known as PLI in Spanish-language optimization practice, with binary decision variables that determine client participation subject to cardinality, latency, carbon-budget, and reliability constraints. For scalability, the implementation includes a greedy surrogate that approximates the MILP objective when exact optimization is unavailable or too expensive.

The contributions of this work are:

- A systematic approach to problem framing for sustainable and resilient FL in the edge-cloud continuum, integrating literature on FL optimization, client selection, compression, green AI, and edge-cloud orchestration.
- A mathematical model and MILP formulation that jointly capture accuracy utility, communication, energy, carbon intensity, latency, reliability, and participation constraints.
- EcoRes-FL, a practical orchestration algorithm that combines carbon-aware selection, reliability filtering, proximal local optimization, and sparse updates.
- A reproducible Python implementation featuring non-IID Dirichlet partitioning, heterogeneous edge/cloud profiles, failure injection, ablation grids, and high-resolution heatmaps.
- An experimental protocol designed to scale from CPU-only quick experiments to larger datasets such as Fashion-MNIST, CIFAR-10, Covertype, and LEAF/FEMNIST.

The rest of the paper is organized as follows. Section II reviews the literature and identifies the gap. Section III defines the system model and formulates the optimization problem. Section IV presents EcoRes-FL. Section V describes the implementation, datasets, metrics, preliminary results, and ablation analysis. Section VI concludes and outlines the path to full-scale validation.

II. Background and Related Work

We conducted a focused systematic literature review to contextualize this work. The search strategy was designed to capture five research streams: FL foundations, heterogeneity-aware optimization, participant selection, communication-efficient FL, and sustainable/carbon-aware edge-cloud computing. Search expressions combined *federated learning*, *edge-cloud continuum*, *client selection*, *stragglers*, *energy-efficient federated learning*, *carbon-aware federated learning*, *green federated learning*, and *model compression*. We prioritized studies indexed in IEEE Xplore, ACM Digital Library, ScienceDirect/Elsevier, PMLR, USENIX, and highly cited arXiv preprints that later became standard references. Inclusion criteria were: (i) explicit relevance to FL systems or edge/cloud orchestration, (ii) methodological contribution or survey value, and (iii) clear implications for sustainability, resilience, communication, or heterogeneity. Exclusion criteria were: (i) centralized-only training, (ii) studies without system or optimization relevance, and (iii) studies in which energy or sustainability was merely motivational and not modeled.

A. Federated Learning Foundations

FL was introduced as a decentralized training paradigm in which clients keep their data local and exchange model updates with a coordinating server. FedAvg remains the dominant baseline because it reduces communication rounds by performing multiple local optimization steps before server aggregation [1]. However, FL differs from conventional distributed optimization because clients are usually statistically heterogeneous, intermittently available, and resource-constrained. Bonawitz *et al.* described how these conditions affect production-scale mobile FL, emphasizing scheduling, secure aggregation, partial participation, and reliability as first-order system concerns [2]. Kairouz *et al.* further consolidated open problems in FL, including privacy, communication, fairness, robustness, personalization, and system heterogeneity [12].

B. Statistical and Systems Heterogeneity

Non-IID data is a central challenge in FL. Local client steps can drift toward client-specific distributions, leading to unstable or slow global convergence. FedProx addresses this by adding a proximal term that discourages local models from moving too far from the global model [3]. SCAFFOLD corrects client drift using control variates and improves convergence under heterogeneous client sampling [4]. Other approaches explore personalization, clustered FL, and adaptive aggregation, but most assume that client participation is determined externally or that sustainability metrics are secondary.

System heterogeneity is equally important. Real clients differ in CPU/GPU capacity, battery state, network bandwidth, and availability. FedCS was an early attempt to select clients in mobile edge computing under resource constraints [13]. Oort later proposed guided participant selection that balances statistical utility and system efficiency, thereby improving time-to-accuracy [5]. More recent work on resource-constrained systems jointly configures client selection and local epochs,

showing that selection and computation cannot be optimized independently [14]. These studies motivate our view that client participation is an optimization problem rather than a random sampling step.

C. Communication Compression and Sparse Updates

Communication is often the dominant bottleneck in FL. Deep Gradient Compression showed that sparse gradient exchange can reduce distributed-training communication by orders of magnitude while preserving accuracy when correction mechanisms are applied [6]. In FL, compression can reduce wireless energy consumption and round duration, but aggressive sparsification may amplify bias under non-IID data. FedGreen for green mobile edge computing uses fine-grained gradient compression to reduce energy consumption while meeting an accuracy requirement [9]. The key insight of our work is that sparsification should not be treated only as a bandwidth technique; it also affects latency, energy, carbon footprint, and resilience to stragglers.

D. Edge-Cloud Continuum and Federated Learning

The edge-cloud continuum connects heterogeneous resources from IoT devices to cloud servers. Prigent *et al.* provide a recent ScienceDirect survey of FL across the computing continuum, identifying system placement, orchestration, heterogeneity, monitoring, and benchmarking limitations as open challenges [15]. Surveys on mobile edge FL and edge intelligence likewise emphasize that FL deployments must account for resource constraints, communication bottlenecks, and privacy requirements [16], [17]. However, many edge FL studies still optimize for latency or energy without explicitly modeling the carbon intensity of computation and communication across geography and time.

E. Sustainable and Carbon-Aware Learning

Green AI research argues that energy use and carbon footprint should be reported alongside accuracy and computational performance. Carbontracker and CodeCarbon are practical tools for tracking or estimating emissions during ML training [7], [18]. Patterson *et al.* show that location, hardware, data-center efficiency, and model design can change ML carbon emissions by large factors [8]. In FL, sustainability introduces additional complexity because local computation, communication, and client geography interact.

Carbon-aware FL has recently emerged. CAFE optimizes FL across geographically distributed data centers under carbon budgets by leveraging carbon-intensity variations and selection strategies [11]. FedGreen adapts model size to clients' carbon profiles and locations [10]. These studies are directly relevant to our work, but they focus primarily on data-center or model-size adaptation settings. In contrast, our framework targets a mixed edge-cloud continuum in which low-resource edge nodes, edge-GPU nodes, and cloud nodes coexist amid failures, stragglers, and non-IID client distributions.

F. Positioning and Research Gap

Table I summarizes representative work. The gap is not the absence of FL, green AI, or client selection individually; it is their joint treatment within a reproducible edge-cloud continuum model. Our contribution serves as a bridge between these lines: it formulates the orchestration problem as a MILP, provides an executable simulator, and jointly evaluates accuracy, macro-F1, communication, energy proxy, carbon proxy, drop rate, and selection behavior.

III. Problem Formulation

This section formalizes the edge-cloud FL setting. We use K clients indexed by $k \in \{1, \dots, K\}$ and T communication rounds indexed by t. Client k stores a private dataset $\mathcal{D}_k$ of size n_k, and the global objective is the standard weighted FL objective:

$$\min_{\mathbf{w}} F(\mathbf{w}) = \sum_{k=1}^{K} p_k F_k(\mathbf{w}), \qquad p_k = \frac{n_k}{\sum_{j=1}^{K} n_j}, \tag{1}$$

where

$$F_k(\mathbf{w}) = \frac{1}{n_k} \sum_{(\mathbf{x},y)\in\mathcal{D}_k} \ell(\mathbf{w}; \mathbf{x}, y) \tag{2}$$

is the empirical loss for client k. In each round, the server selects a subset $\mathcal{S}_t$ of clients. Each selected client performs local training, and the server aggregates the local updates.

A. Edge-Cloud Client Model

Each client is characterized by a systems profile:

$$\pi_k = (q_k, b_k, e_k^{cmp}, e_k^{net}, r_k, \gamma_{k,t}), \tag{3}$$

where q_k is compute capacity, b_k is uplink/downlink bandwidth, e_k^{cmp} is the computation energy coefficient, e_k^{net} is the network energy coefficient, $r_k \in [0, 1]$ is the reliability, and $\gamma_{k,t}$ is the carbon intensity at round t in gCO_2eq/kWh. This profile can represent low-power edge nodes, mid-range gateways, edge-GPU nodes, and cloud nodes.

Let $s_t \in (0, 1]$ denote the update density after compression, where $s_t = 1$ is dense communication and smaller values represent top-k sparse updates. Let P be the number of model parameters and assume 32-bit floating-point updates. The approximate communication volume for client k at round t is:

$$M_{k,t}(s_t) = 2 \cdot 4Ps_t/2^{20}, \tag{4}$$

measured in MB, where the factor 2 approximates upload and amortized broadcast. The local compute time and network time are modeled as:

$$\tau_{k,t}^{cmp} = \frac{\rho n_k P E_t}{q_k}, \qquad \tau_{k,t}^{net} = \frac{8M_{k,t}(s_t)}{b_k}, \tag{5}$$

where E_t is the number of local epochs and ρ is a proportionality constant. Total round contribution time is:

$$\tau_{k,t} = \tau_{k,t}^{cmp} + \tau_{k,t}^{net}. \tag{6}$$

TABLE I: Positioning of representative literature and the gap addressed by this work.

Work	Main focus	Non-IID	Client selection	Compression	Carbon / sustainability	Edge-cloud continuum
FedAvg [1]	FL baseline	Yes	Random	No	No	Limited
FedProx [3]	Heterogeneity	Yes	Partial	No	No	Limited
SCAFFOLD [4]	Client drift	Yes	Sampling-aware	No	No	Limited
FedCS [13]	MEC selection	Partial	Yes	No	Energy-related	Edge-oriented
Oort [5]	Utility/system selection	Yes	Yes	No	No	Device-scale
DGC [6]	Communication	Partial	No	Yes	Indirect	No
FedGreen-MEC [9]	Green MEC FL	Yes	Partial	Yes	Energy-focused	Edge-oriented
CAFE [11]	Carbon-aware FL	Yes	Yes	No	Explicit carbon	Geo-distributed cloud
FedGreen-size [10]	Carbon/model size	Yes	Partial	Model-size adaptation	Explicit carbon	Cloud/edge clients
Continuum survey [15]	Systems survey	Yes	Discussed	Discussed	Discussed	Explicit
Our Proposal (EcoRes-FL)	Sustainable resilient FL	Yes	MILP/greedy	Top-k sparse	Explicit carbon proxy	Mixed edge-cloud

The energy proxy is:

$$\mathcal{E}_{k,t} = e_k^{cmp} n_k PE_t + e_k^{net} M_{k,t}(s_t), \tag{7}$$

and the operational carbon proxy is:

$$\mathcal{C}_{k,t} = \frac{\mathcal{E}_{k,t}}{3.6 \times 10^6} \gamma_{k,t}. \tag{8}$$

The denominator converts Joules to kWh.

B. Statistical Utility

Let $u_{k,t}$ denote the expected learning utility of client k at round t. In a practical system, $u_{k,t}$ cannot depend on private raw data held by other clients. It can be estimated using metadata-compatible proxies such as local dataset size, label entropy under privacy-preserving reporting, validation improvement from previous contributions, gradient novelty, or stale-participation penalties. The implementation uses a transparent proxy based on local data size and system feasibility, while the mathematical formulation allows more sophisticated utility estimators:

$$u_{k,t} = \phi(n_k, h_k, \Delta_{k,t-1}, a_{k,t}), \tag{9}$$

where h_k is a local heterogeneity descriptor, $\Delta_{k,t-1}$ is the previous validation utility, and $a_{k,t}$ captures client availability.

C. Mixed-Integer Linear Programming Formulation

We define a binary decision variable $x_{k,t} \in \{0,1\}$, where $x_{k,t} = 1$ if client k is selected in round t. For a fixed compression density s_t and a local epoch configuration, client costs are constants before scheduling. The round-level client-selection problem can be formulated as the following mixed-integer linear program:

$$\max_{\mathbf{x}_t} \quad \sum_{k=1}^{K} x_{k,t} \left(\alpha_u u_{k,t} - \alpha_c \widehat{\mathcal{C}}_{k,t} - \alpha_m \widehat{M}_{k,t} - \alpha_\tau \widehat{\tau}_{k,t} + \alpha_r r_k \right) \tag{10}$$

$$\text{s.t.} \quad m_{\min} \le \sum_{k=1}^{K} x_{k,t} \le m_{\max}, \tag{11}$$

$$\sum_{k=1}^{K} x_{k,t} \mathcal{C}_{k,t} \le B_t^{CO_2}, \tag{12}$$

$$\sum_{k=1}^{K} x_{k,t} M_{k,t} \le B_t^{net}, \tag{13}$$

$$x_{k,t} \tau_{k,t} \le \tau_t^{max}, \quad \forall k, \tag{14}$$

$$\sum_{k=1}^{K} x_{k,t} r_k \ge R_t^{min} \sum_{k=1}^{K} x_{k,t}, \tag{15}$$

$$\sum_{k \in \mathcal{G}_g} x_{k,t} \ge L_{g,t}, \quad \forall g \in \mathcal{G}, \tag{16}$$

$$x_{k,t} \in \{0,1\}, \quad \forall k. \tag{17}$$

Here, $\widehat{\mathcal{C}}_{k,t}$, $\widehat{M}_{k,t}$, and $\widehat{\tau}_{k,t}$ are normalized costs; $B_t^{CO_2}$ is the round carbon budget; B_t^{net} is the communication budget; τ_t^{max} is a latency deadline; R_t^{min} is the minimum average reliability; and $\mathcal{G}$ denotes optional coverage groups, such as regions, device tiers, or label-distribution strata. Constraint (16) prevents a carbon-aware policy from repeatedly selecting only a narrow set of clients, thereby worsening statistical bias.

This formulation is intentionally modular. If the system has no explicit carbon budget, Constraint (12) can be relaxed, and carbon can remain in the objective. If exact MILP solving is too expensive for large K, the objective coefficients provide a greedy ranking surrogate:

$$score_{k,t} = \alpha_u u_{k,t} - \alpha_c \widehat{\mathcal{C}}_{k,t} - \alpha_m \widehat{M}_{k,t} - \alpha_\tau \widehat{\tau}_{k,t} + \alpha_r r_k. \tag{18}$$

The implementation includes both an MILP function and a greedy fallback.

D. Optimization Interpretation

The formulation is multi-objective. Accuracy utility is maximized while communication, latency, energy, and carbon are minimized. Rather than collapsing sustainability into a vague environmental motivation, the model exposes the trade-off directly through α_c and $B_t^{CO_2}$. A high value of α_c favors low-carbon clients, but may reduce statistical diversity if low-carbon clients are concentrated in a subset of the data distribution. A high value of α_u prioritizes learning progress but may select high-carbon or unreliable clients. The ablation study in Section V is designed to observe these trade-offs across different client counts, non-IID severity, sparsity levels, and failure rates.

E. Research Questions

The paper is guided by the following research questions:

- **RQ1:** Can a sustainability- and reliability-aware client-selection policy reduce communication and carbon proxy without severe accuracy degradation?
- **RQ2:** How sensitive is the approach to client scale and to non-IID severity?
- **RQ3:** How does sparse communication interact with client failures and stragglers?
- **RQ4:** Which visual diagnostics help determine whether carbon-aware selection introduces harmful participation bias?

IV. Methodology

EcoRes-FL is a sustainability- and resilience-aware FL orchestration framework. It does not replace the learning model; rather, it controls which clients participate, how local objectives are regularized, and how much of the update is transmitted. Fig. 1 summarizes the workflow. The server monitors a lightweight profile for each client, computes a carbon-aware participation score or solves the MILP, dispatches the global model to selected clients, receives sparse local updates, filters failed contributions, and aggregates valid updates.

A. Local Objective

For each selected client k, EcoRes-FL uses a proximal local objective:

$$\min_{\mathbf{w}} F_k(\mathbf{w}) + \frac{\mu}{2}\|\mathbf{w} - \mathbf{w}_t\|_2^2, \tag{19}$$

where $\mathbf{w}_t$ is the current global model and $\mu \geq 0$ controls how strongly the local model is regularized toward the global state. When $\mu = 0$, the local step reduces to FedAvg-style training. The proximal term is useful when clients are statistically heterogeneous or when they perform different amounts of local work.

B. Sparse Update Transmission

After local training, client k computes $\Delta_{k,t} = \mathbf{w}_{k,t}^{local} - \mathbf{w}_t$. EcoRes-FL keeps only the entries with the largest magnitudes:

$$\mathcal{T}_s(\Delta_{k,t})_i = \begin{cases} \Delta_{k,t,i}, & |\Delta_{k,t,i}| \geq \theta_s, \\ 0, & \text{otherwise}, \end{cases} \tag{20}$$

Algorithm 1: EcoRes-FL: Sustainable and Resilient Federated Learning

Input: Clients $\{1, \ldots, K\}$, rounds T, model $\mathbf{w}_0$, density s, local epochs E, proximal coefficient μ, deadline τ^{max}

Output: Global model $\mathbf{w}_T$

Initialize monitoring table with $q_k, b_k, r_k, \gamma_{k,t}, n_k$;
for $t = 0$ **to** $T - 1$ **do**
  Estimate $u_{k,t}$, $\mathcal{C}_{k,t}$, $M_{k,t}$, $\tau_{k,t}$ for all available clients;
  Select $\mathcal{S}_t$ by solving the MILP in Eq. (10)–(16); if unavailable, use greedy score;
  foreach $k \in \mathcal{S}_t$ *in parallel* **do**
    Receive $\mathbf{w}_t$ from server;
    Train local model using $F_k(\mathbf{w}) + \frac{\mu}{2}\|\mathbf{w} - \mathbf{w}_t\|^2$;
    Compute update $\Delta_{k,t}$;
    Send sparse update $\mathcal{T}_s(\Delta_{k,t})$;
  Filter failed clients and stragglers to obtain $\mathcal{V}_t$;
  if $|\mathcal{V}_t| > 0$ **then**
    Aggregate weighted sparse updates to obtain $\mathbf{w}_{t+1}$;
  else
    Set $\mathbf{w}_{t+1} \leftarrow \mathbf{w}_t$;
  Log accuracy, macro-F1, communication, energy proxy, carbon proxy, drop rate, and selected-client heatmap;
return $\mathbf{w}_T$;

where θ_s is chosen so that approximately a density s of entries remains. This top-k mechanism reduces update size, transmission energy, and the probability of missing the deadline. In future versions, error feedback can be added to compensate for dropped entries; the current implementation uses direct top-k sparsification to keep the baseline transparent.

C. Aggregation

Let $\mathcal{V}_t \subseteq \mathcal{S}_t$ be the set of selected clients that successfully submit updates before the accepted deadline. The server aggregates:

$$\mathbf{w}_{t+1} = \mathbf{w}_t + \sum_{k \in \mathcal{V}_t} \frac{n_k}{\sum_{j \in \mathcal{V}_t} n_j} \mathcal{T}_s(\Delta_{k,t}). \tag{21}$$

This preserves the sample-size weighting of FedAvg while supporting sparse updates. If $\mathcal{V}_t$ is empty, the server keeps $\mathbf{w}_{t+1} = \mathbf{w}_t$ and logs a failed round.

D. Algorithm

Algorithm 1 outlines the complete procedure. The method is intentionally compatible with multiple local learners. The released code uses a fast softmax classifier for reproducibility and includes hooks for PyTorch datasets.

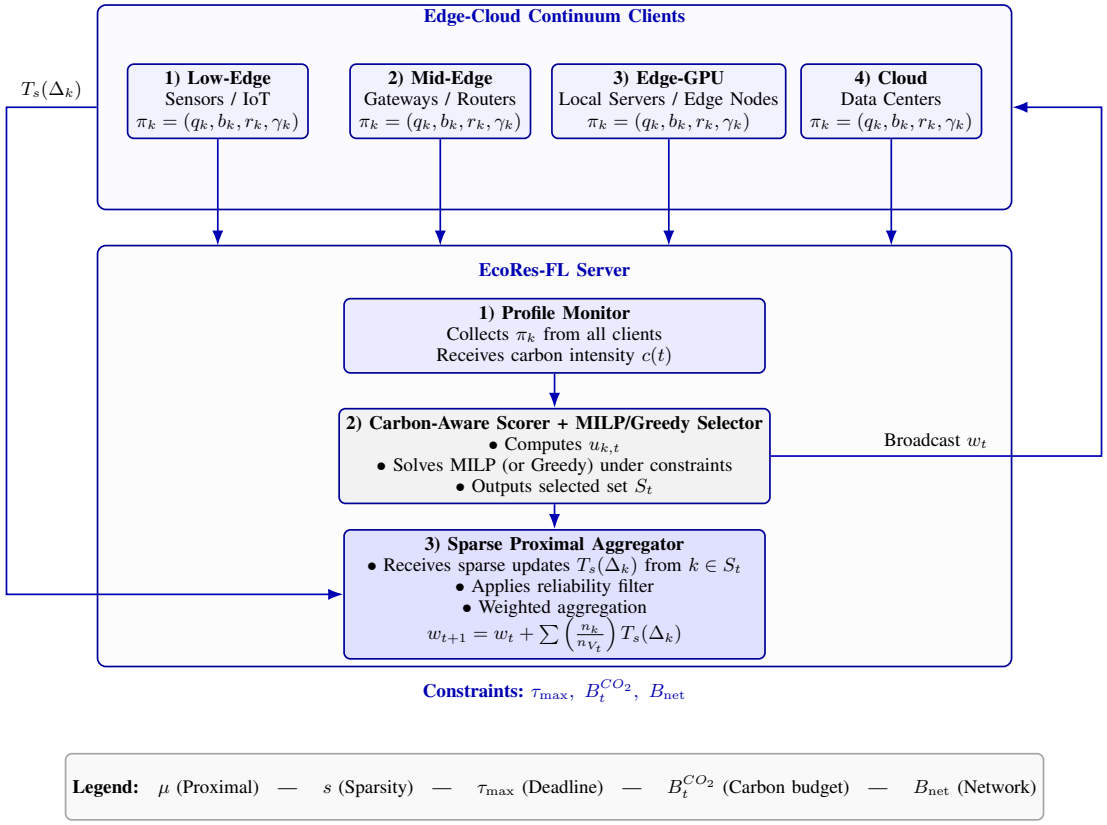


Fig. 1: High-level architecture of EcoRes-FL for sustainable and resilient FL in the edge-cloud continuum.

E. Implementation Design

To support future research, we make our approach publicly available on https://github.com/alexandercooper97/EcoRes-FL. It includes:

- Dirichlet non-IID partitioning controlled by parameter α.
- Heterogeneous client profiles for low-edge, mid-edge, edge-GPU, and cloud tiers.
- Carbon-intensity traces simulated as time-varying profiles; these can be replaced by real traces from Electricity Maps, WattTime, or regional grid datasets.
- Failure and straggler injection using reliability and latency models.
- Baselines: FedAvg, FedProx, CarbonAware-FL, and EcoRes-FL.
- Ablation grids over number of clients, non-IID severity, sparsity, and failure probability.
- Publication-quality figures with 600-dpi PNG output and font settings compatible with IEEE PDF workflows.

F. Why This Design is Scientifically Useful

The framework's main purpose is not to claim that any scheduler dominates across all conditions. Instead, it enables controlled experimentation across competing objectives. A random scheduler may achieve high accuracy but waste communication. A carbon-only scheduler may select low-carbon clients but reduce statistical diversity. A compression-heavy scheduler may reduce communication but harm convergence. EcoRes-FL is designed to make these trade-offs explicit, so that a study can report not only final accuracy but also the sustainability-resilience envelope in which a method is beneficial.

V. Experimental Evaluation

A. Proof-of-Concept Simulation Setup

The experimental evaluation consists of a CPU-only, reproducible proof-of-concept (quick protocol) that validates the pipeline in a controlled edge-cloud simulator. This simulation does not involve real distributed devices; instead, it models K clients with heterogeneous profiles, synthetic carbon-intensity traces, and simulated failures. The purpose is to verify that the proposed metrics behave as expected and to expose the accuracy–sustainability trade-off before any physical deployment. All datasets can be generated or loaded without external downloads, ensuring full reproducibility.

The quick protocol is a CPU-only, self-contained simulation that validates the scoring, selection, and sparsification pipeline under controlled conditions. Its purpose is to verify that the accuracy, carbon-proxy, and communication metrics behave consistently, and to generate comparative diagnostics and ablation heatmaps. Results obtained with this protocol represent a proof-of-concept and should be interpreted as indicative trends rather than absolute measurements. A multi-seed validation on larger benchmarks is outlined as future work in Section VI.

TABLE II: Datasets supported by the experimental framework.

Dataset	Experim. mode	Classes	Purpose
Synthetic-Vision [19]	Quick	10	Noisy high-dimensional multi-class proxy for edge vision workloads.
IoT-Fault [20]	Quick	5	Imbalanced condition-monitoring proxy for resilient edge analytics.
Digits [21]	Quick	10	Sanity check for low-resource reproducibility.
Fashion-MNIST [22]	Full	10	More challenging image classification with non-IID client partitions.
CIFAR-10 [23]	Full	10	Color image classification; useful for CNN or embedding-based FL.
Covertype [24]	Full	7	Large tabular dataset for geographically distributed environmental sensing.
LEAF/FEMNIST [25]	Full	62	Realistic federated benchmark with naturally partitioned writers.

B. Baselines

We compare four methods on three datasets (Synthetic Vision, IoT-Fault, and Digits), as described in table II:

- **FedAvg**: random client selection, dense updates, and no proximal regularization.
- **FedProx**: random client selection, dense updates, and proximal local training.
- **CarbonAware-FL**: carbon- and latency-aware client ranking with dense updates.
- **EcoRes-FL**: carbon/reliability-aware selection, proximal local training, and sparse top-k updates.

The metrics are final accuracy, macro-F1, cumulative communication in MB, energy proxy in Joules, carbon proxy in gCO_2eq, drop rate, and contribution rate.

C. Reproducibility Settings

Unless otherwise stated, the preliminary run uses 24 clients, 8 clients per round, 12 rounds, a Dirichlet non-IID parameter $\alpha = 0.25$, two local epochs, a failure probability of 0.08, and a top-k density of $s = 0.30$ for EcoRes-FL. Client profiles are sampled from four tiers: low-edge, mid-edge, edge-GPU, and cloud. Carbon intensity varies across clients and rounds according to a sinusoidal-plus-noise profile.

D. Preliminary Results

Table III reports the CPU-only preliminary results generated by the included code. On Synthetic-Vision, EcoRes-FL substantially reduces communication relative to dense baselines but incurs a small accuracy loss; this is expected because the dataset is noisy and the sparse density is aggressive. On IoT-Fault, EcoRes-FL improves final accuracy relative to FedAvg while using much less communication. On Digits, EcoRes-FL keeps competitive accuracy, but its carbon proxy is not lower than all dense methods because the selected low-latency clients can exhibit higher carbon intensity in the simulated trace. This highlights the need for a study to report the full trade-off rather than a single scalar metric.

Fig. 2 shows convergence for the high-dimensional Synthetic-Vision proxy. The graphic helps determine whether sparse updates delay early learning. Fig. 3 shows the accuracy–carbon trade-off across datasets and methods. These plots are more informative than a single final score because a sustainable FL method should be evaluated as a Pareto optimization problem.

The execution results of the proposed framework using the quick protocol are available on a public simulator https://alexandercooper97.github.io/EcoRes-FL

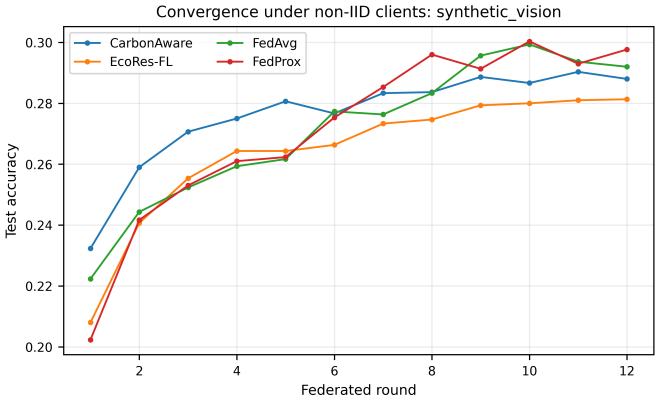


Fig. 2: Convergence under non-IID clients on Synthetic-Vision.

E. Heatmap-Based Ablation Study

The framework generates heatmaps to analyze parameter sensitivity. Fig. 4 varies the number of clients and the Dirichlet non-IID parameter α. Smaller α values yield more heterogeneous client distributions. This visualization helps assess whether the scheduler remains stable as client data becomes more skewed. Fig. 5 varies sparse density and failure probability. This is important because sparse updates reduce communication but can make learning more fragile if failures remove too many informative contributions.

Fig. 6 visualizes selected clients by round. This diagnostic is essential for a carbon-aware scheduler: if only a few clients are repeatedly selected, the method may reduce carbon while harming fairness and statistical coverage. A future version should add fairness constraints or minimum group coverage constraints directly into Eq. (16).

F. Threats to Validity

The current preliminary implementation uses proxy energy and carbon models rather than hardware-level measurements. Therefore, the carbon results should be interpreted as relative trends rather than absolute emissions. The quick datasets are useful for reproducibility but do not replace real federated benchmarks. The top-k sparse update does not yet implement error feedback, which may underestimate the accuracy of advanced compression methods.

TABLE III: Preliminary CPU-only results. These are single-seed results produced by the included simulator and should be extended to multi-seed experiments.

Dataset	Method	Acc.	Macro-F1	Comm. MB	Energy J	Carbon g	Drop rate
Synthetic-Vision	FedAvg	0.292	0.279	0.768	2241.48	0.155	0.188
	FedProx	0.298	0.292	0.866	2403.05	0.159	0.083
	CarbonAware-FL	0.288	0.278	0.896	2203.35	0.172	0.052
	EcoRes-FL	0.281	0.268	0.260	2123.63	0.168	0.083
IoT-Fault	FedAvg	0.540	0.210	0.231	1175.99	0.093	0.031
	FedProx	0.547	0.233	0.221	1195.05	0.086	0.073
	CarbonAware-FL	0.546	0.226	0.223	2018.60	0.161	0.063
	EcoRes-FL	0.553	0.245	0.063	1900.96	0.149	0.125
Digits	FedAvg	0.860	0.854	0.426	137.89	0.010	0.104
	FedProx	0.849	0.838	0.436	143.16	0.011	0.083
	CarbonAware-FL	0.858	0.854	0.422	119.09	0.011	0.115
	EcoRes-FL	0.838	0.833	0.131	125.12	0.012	0.083

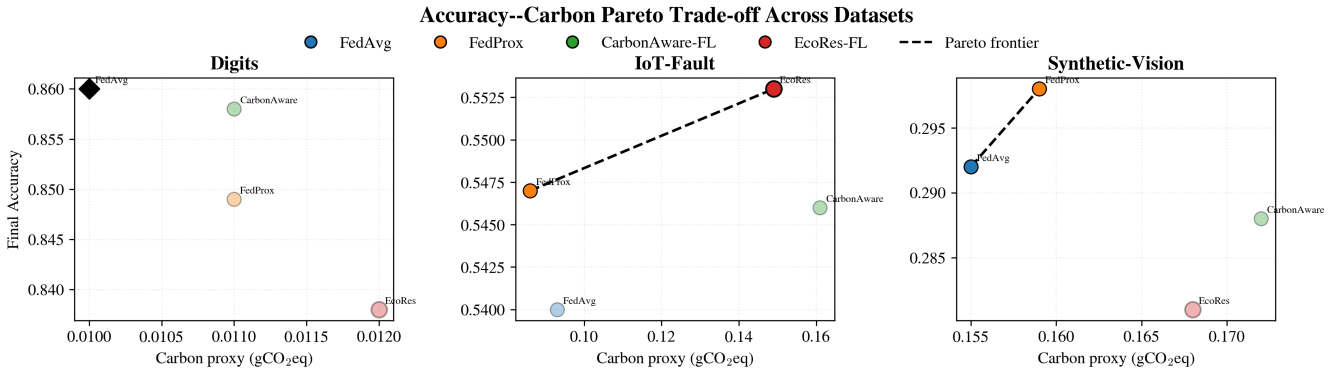
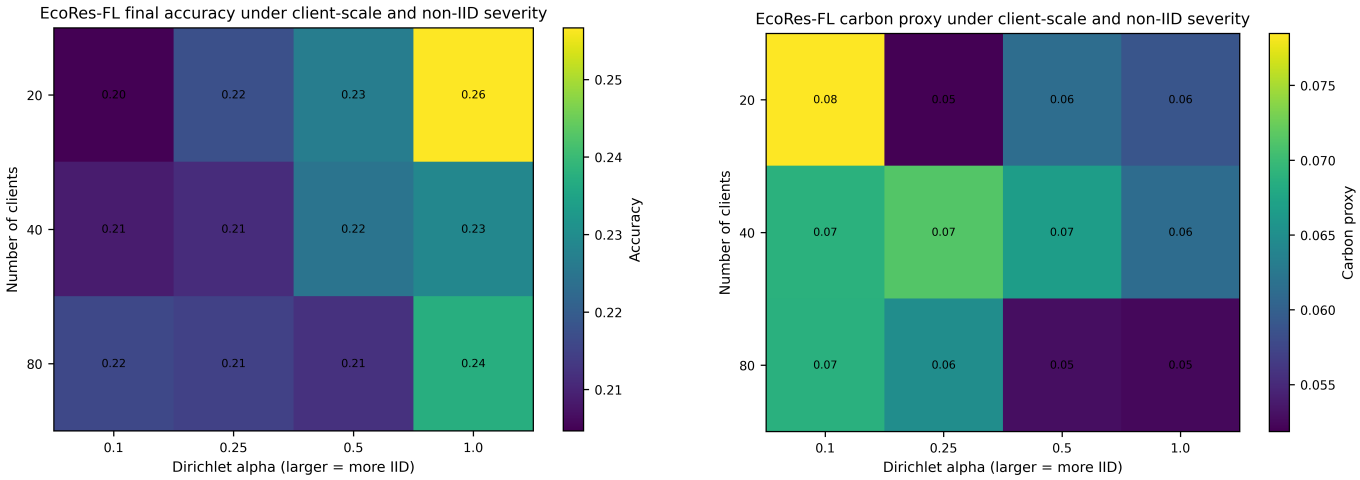


Fig. 3: Accuracy–carbon Pareto trade-off across datasets.



(a) Accuracy vs. clients and non-IID severity.

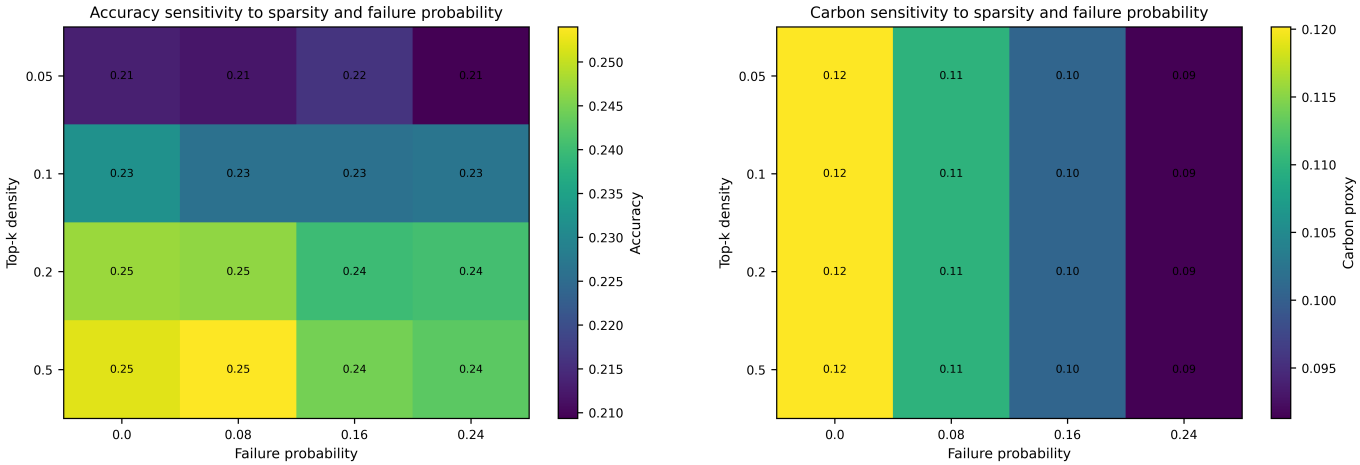
(b) Carbon proxy vs. clients and non-IID severity.

Fig. 4: Client-scale and non-IID ablation for EcoRes-FL.

The MILP formulation is exact for a fixed round with precomputed linear costs, but real systems may have nonlinear effects such as queueing, shared wireless interference, and time-varying availability. These limitations are not weaknesses of the research direction; they define the next steps for validation.

VI. Conclusion and Future Work

This paper presented EcoRes-FL, a sustainability- and resilience-aware orchestration framework that formulates client participation as a mixed-integer linear program and combines carbon-aware scoring, proximal local training, and



(a) Accuracy sensitivity.

(b) Carbon sensitivity.

Fig. 5: Ablation over sparse-update density and failure probability.

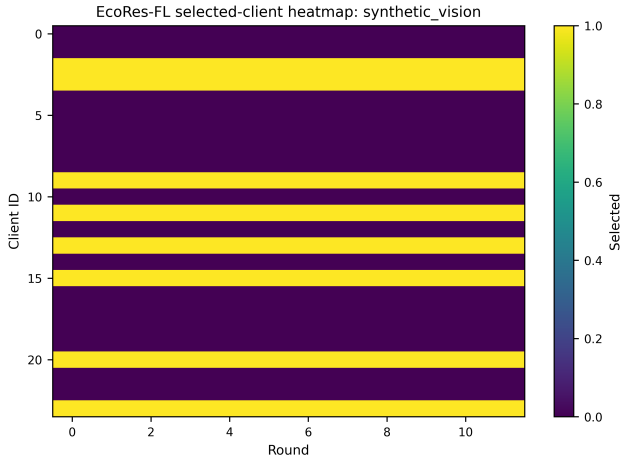


Fig. 6: Selected-client heatmap for EcoRes-FL on Synthetic-Vision.

top-k sparse updates. A reproducible CPU-only simulator was developed to evaluate the framework under heterogeneous edge-cloud profiles, non-IID data, and simulated failures.

The simulation-based experiments indicate that EcoRes-FL substantially reduces communication overhead and, in most evaluated scenarios, improves the carbon proxy metric relative to dense baselines while maintaining competitive predictive performance. The heatmaps further show that the framework remains stable under moderate levels of non-IID severity and sparsity, although extreme values of either parameter lead to noticeable performance degradation.

Future work will focus on five directions. First, we will run multi-seed experiments and report confidence intervals. Second, we will integrate real carbon-intensity traces from public grid datasets or carbon APIs. Third, we will add SCAFFOLD, Oort-style participant selection, and error-feedback compression as stronger baselines. Fourth, we will extend the MILP with fairness and coverage constraints to prevent repeatedly selecting only low-carbon but statistically narrow clients. Finally, we will validate the framework on a physical or emulated edge-cloud testbed, where measured energy, latency, and failure traces can replace proxy models.

To further assess the generality of these findings, the framework will be evaluated on larger-scale benchmarks, including Fashion-MNIST, CIFAR-10, Covertype, and the naturally partitioned LEAF/FEMNIST dataset, with client populations ranging from 50 to 200 and at least five random seeds per configuration. These experiments will produce a full Pareto analysis of the sustainability-resilience envelope and will help establish statistical confidence intervals for the trade-offs reported here.

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